How-To Library

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Running a Profile on Wide Tables

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Abstract

You might need to perform discovery tasks on a relational data source with wide tables or flat files with a huge number of columns. To optimize the profile run on wide tables, you can configure some of the Data Integration Service parameters specific to Profiling Service Module. You can run a column profile on the relational source. Column profile results help you identify and exclude columns that do not meet the requirements of a key column based on profiling statistics. You can then perform primary key discovery to discover the relevant primary keys. Finally, perform a foreign key discovery to identify all relevant relationships between multiple tables in the data source.

Supported Versions

- Data Explorer 9.5.1
- Data Quality 9.5.1 - 9.6.1
- Data Services 9.5.1 - 9.6.1

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Overview

Wide tables in the data source might have the number of columns in the range of 200 to 500. Occasionally, you might come across wide tables with more than 1000 columns. Before you run a profile on wide tables, you can configure a few Data Integration Service parameters for optimal performance.

You can configure the following Data Integration Service parameters that can affect the profiling performance on wide tables:

- Maximum Execution Pool Size
- Maximum Profile Execution Pool Size
- Maximum DB Connections
- Maximum Heap Size
- Maximum Concurrent Profile Threads

Profile Configuration in Data Integration Service

The Data integration Service has a set of parameters specific to the Profiling Service Module that controls the performance of a profiling job. You must configure these parameters for the Data Integration Service.

You can configure the following types of parameters:

Profiling warehouse database properties

Constitute the majority of the parameters and apply to the Profiling Service Module across the profile deployment in the organization. For example, the Maximum Profile Execution Pool Size parameter controls the total number of mappings that the Profiling Service Module generates. The profiling warehouse database properties apply to all the nodes that constitute the Profiling Service Module.

Advanced profiling properties

Apply to a single Data Integration Service node. You must configure the advanced profiling parameters for each node in the Data Integration Service if you use a grid.

You can use the following sections in the Administrator tool to configure the parameters for the Data Integration Service:

- Profiling warehouse database properties on the **Properties** tab.
- Advanced profiling properties on the **Properties** tab.

Profiling Warehouse Database Properties

The profiling warehouse database properties apply to the Profiling Service Module across the deployment.

You can set the following parameters:

Profiling Warehouse Database

Connection name to the profiling warehouse database. In addition to the profile results, the profiling warehouse holds the persisted profile job queue. Verify that no profile job is running when you change the connection name. If profile jobs are running when you change the connection name, the jobs might stop running because the profile jobs run on the Data Integration Service where the Profiling Service Module submitted the profile jobs.

You set the default value when you create the Data Integration Service instance.

Maximum Ranks

The number of minimum values and maximum values to display for a profile based on the data type of the column. Ranks are useful to understand the range that the values of a column might take and whether the column has false null values. You can retain the default value.

Default is 5.

Maximum Patterns

The maximum number of patterns that each column stores. Sometimes, it is important to store as many patterns as possible. You can set the **Maximum Patterns** parameter to a large number, such as 10,000, and adjust the **Pattern Threshold Percentage** parameter to **0.01**. Setting a high value for this parameter has negligible impact on performance.

Default is 10.

Maximum Profile Execution Pool Size

The number of profile mappings that the Profiling Service Module can run concurrently when the Data Integration Service runs on a single node or on a grid. The pool size is dependent on the aggregate processing capacity of the Data Integration Service, which you specify for each node on the **Processes** tab of the Administrator tool. The pool size cannot be greater than the aggregate value of the processing capacity of all nodes.

When you plan for a deployment, consider the following types of threads:

- Threads used for profile tasks.

- Threads reserved for drill-down tasks on the source data and similar, quick real-time profile tasks.
- Threads for all other nonprofiling purposes, such as SQL endpoints, preview, and deployed mappings.

To effectively configure the Maximum Execution Pool Size parameter, you must understand the combination of mappings jobs and profile jobs. For optimal performance, verify that the total number of threads in the thread categories adds up to the value of the Maximum Execution Pool Size parameter.

Default is 10.

Maximum DB Connections

The number of parallel queries across all the profiling jobs for a relational source. If two profile jobs run on the same database, each profile job gets half the number of connections. If two profile jobs run on different databases, each profile job gets the maximum number of connections. If the relational database that contains the wide table has sufficient processing capabilities, you can increase this value so that a column profile concurrently runs on more columns. The concurrent profile run decreases the overall profile run time. A general guideline is to associate one CPU core on the relational database machine to one database connection.

Note: You can run profiles on multiple wide tables concurrently, and each profile uses the maximum number of connections.

The Profiling Service Module verifies the connection name to recognize the different databases. If you have different connection names, the Profiling Service Module considers the connection names as different databases. Therefore, it is recommended that you do not create database aliases with two different connection names.

All databases use the same number of connections. Therefore, if you run a profile that runs on two databases with different performance characteristics, consider the database with the lowest concurrent profile run requests to configure this parameter.

Default is 5.

Profile Results Export Path

The default path where the Profiling Service Module stores the exported objects, such as Microsoft Excel spreadsheets and DDL files.

The default value is `<Installation Directory>/tomcat/bin/ProfileExport`.

Advanced Profiling Properties

The advanced profiling properties apply to a single Data Integration Service node. You must configure the parameters for each node in the Data Integration Service.

You can configure the following advanced profiling properties:

Pattern Threshold Percent

The minimum percentage of rows matching up to two decimal places for a pattern to appear in the results.

Default is 5.00.

Maximum # Value Frequency Pairs

The maximum number of value frequency pairs stored in the profiling warehouse. This parameter does not control whether the Profiling Service Module computes all the value frequency pairs and the basic characteristics of a column profile run. However, an increase in the parameter value adds additional time to write more value frequency pairs to the profiling warehouse.

Default is 16,000.

Maximum String Length

The maximum length of a string that the Profiling Service Module mappings process internally. The default is set to the maximum value of 255. If you decrease the value, the Data Integration Service truncates the value. Decreased string lengths can have a minor impact on the amount of tablespace required for the profiling warehouse and negligible impact on the overall performance.

Default is 255.

Maximum Numeric Precision

The maximum precision, which is the number of significant digits in the number, for numeric decimal data types. If you set a low value for this parameter, the Data Integration Service might process additional numeric data types as strings instead of numbers.

Default is 38.

Maximum Concurrent Profile Jobs

The number of profile jobs that can run in parallel even if there are more threads available to run mappings. You can use the parameter to control the number of concurrent jobs. Use the parameter to optimize the Profiling Service Module resources so that the Profiling Service Module resources do not affect the resource usage for the Data Integration Service.

When you run a column profile on a relational source, the Maximum DB Connections parameter determines the number of mappings that the Profiling Service Module uses. The Profiling Service Module uses one mapping each for the other profile jobs. When you run a column profile on flat file sources, the Maximum Concurrent Profile Threads parameter determines the number of mappings that the Profiling Service Module uses.

You can configure the Maximum Concurrent Profile Jobs parameter based on the capabilities and other uses of the nodes that you run the Profiling Service Module on.

Default is 5.

Maximum Concurrent Columns

The number of columns that a mapping runs in parallel. The default value of 5 is optimal for most of the profiling deployments. You can increase the default value for columns with cardinality lower than the average value. Decrease the default value for columns with cardinality higher than the average value. You might also want to decrease this value is when you consistently run profiles on large source files where temporary disk space is low.

Default is 5.

Maximum Concurrent Profile Threads

The number of mappings that run in parallel when you run a column profile on a flat file data source. Each mapping simultaneously runs a profile on a number of columns equal to the value you set for the Maximum Concurrent Columns parameter.

If you increase this parameter value, the Profiling Service Module runs the specified number of mappings in parallel and reduces the overall time to run the profile. If you run a column profile or data domain discovery on a wide table, you can increase this value. It is recommended that you add more temporary disks to each Data Integration Service node to balance the additional Input/Output operations required to run a profile on wide tables.

Default is 1.

Maximum Column Heap Size

The cache size for each column profile mapping for flat files. You can increase this value to prevent the Data Integration Service from writing some parts of the intermediate profile results to the temporary disk. This

setting does not apply to large data sources. The default setting is optimal for most of the profiling deployments.

Default is 64.

Reserved Profile Threads

The number of threads that the Data Integration Service reserves to perform drill-down operations. The parameter ensures that a thread is always available for a drill-down operation, which is a quick and real-time operation. If enterprise discovery is a large part of the profile jobs, you can increase the parameter value. You can also increase the parameter value if multiple users might perform drill-down operations when the Profiling Service Module runs a profile.

Default is 1.

For more information on profile sizing, refer https://communities.informatica.com/docs/DOC-9757

Profile Sizing Example on Wide Tables

This example describes the settings for a wide table data source with multiple tables. The tables might have up to 1000 columns on which you need to perform data discovery tasks.

The following table describes the setup environment:

Setup Component	Description
Users	One or two data analysts use the Analyst tool.
Hardware	1 node, 4 cores, 32 GB RAM, and 500 GB temporary disk space.
Data	A relational database that contains multiple tables with more than 1000 columns.
Profile type	Column profile, primary key discovery, and foreign key discovery.
Profiling warehouse	Set up on the same server.
Model Repository Service	Set up on the same server.
Analyst Service	Set up on the same server.

The following table summarizes the recommended configuration parameters for the setup environment:

Parameter	Value
Maximum Execution Pool Size	18
Maximum Profile Execution Pool Size	8
Maximum DB Connections	8
Maximum Heap Size	4096m **Note:** The value is based on the assumption that the server runs only profiling tasks.

Profiling Recommendation

First, run a column profile on the wide-table source. You can use the column profile results to exclude columns that are not likely primary keys. Examples of columns to exclude are columns with many null values and columns that contain more descriptive data than key data.

Next, perform primary key discovery on the data source. Wide tables can trigger a large number of false primary keys that can increase resource usage and run time. Perform primary key discovery as a single data object profile on the wide table to exclude the false primary keys. Then, perform a foreign key discovery on all columns of multiple tables to discover foreign keys and complete the profiling process with better performance and accurate results.

Note: If you perform foreign key discovery without first performing primary key discovery, the Developer tool uses only documented primary keys to identify the foreign key relationships.

Wide Table Guidelines

Verify that you configure the basic settings for profiling before you can fine tune some of the parameters to improve the profiling performance on wide tables.

When you perform data discovery on wide tables, you can consider the following order of profiling operations on the source:

1. Run a column profile to understand the content and quality of each column.
2. Review the structure of the data source. Curate the profile results to exclude columns that are not likely primary keys. You need to restrict the number of primary key columns to a manageable limit. This step helps you increase the performance by reducing the amount of processing required for primary key discovery and foreign key discovery.
3. Perform primary key discovery as a single data object profile.
4. Curate the primary keys of all tables.
5. Perform foreign key discovery.

Guidelines can vary based on the configuration capacity of the Data Integration Service server machine and database. Before you run a column profile on wide tables with 200 to 500 columns, you can configure a few Data Integration Service properties specific to Profiling Service Module.

The following table describes the Data Integration Service properties and the recommended values for multiple combinations of the number of CPU cores and data source types:

Parameter	8 Cores and Flat File	16 Cores and Flat File	8 Cores and Relational Source	16 Cores and Relational Source
Maximum Execution Pool Size	18	26	22	34
Maximum Profile Execution Pool Size	8	16	12	24

Parameter	8 Cores and Flat File	16 Cores and Flat File	8 Cores and Relational Source	16 Cores and Relational Source
Maximum DB Connections	5	5	8	16
Maximum Concurrent Profile Threads	2	3	2	2

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